

UniTS Anomaly Ratio Sweep

STPSat4-TCS | Generated 2026-03-12 10:53

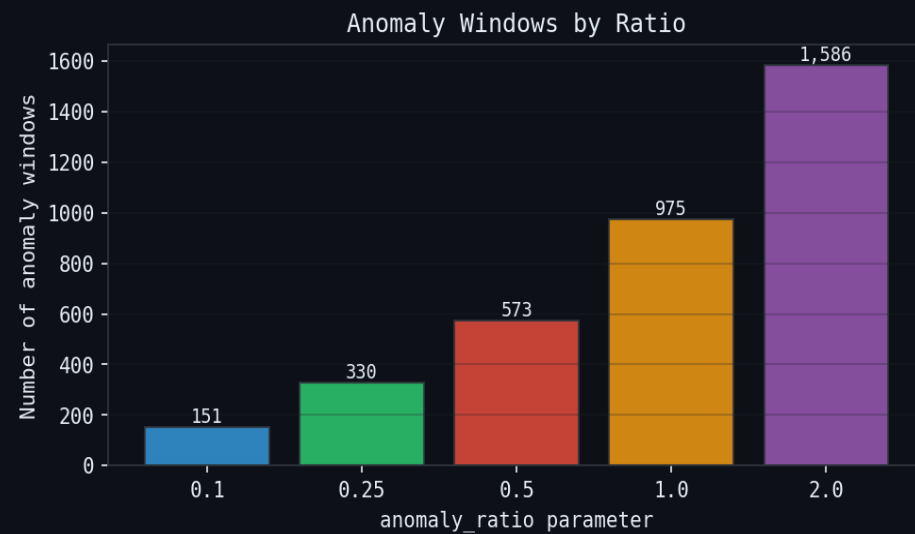
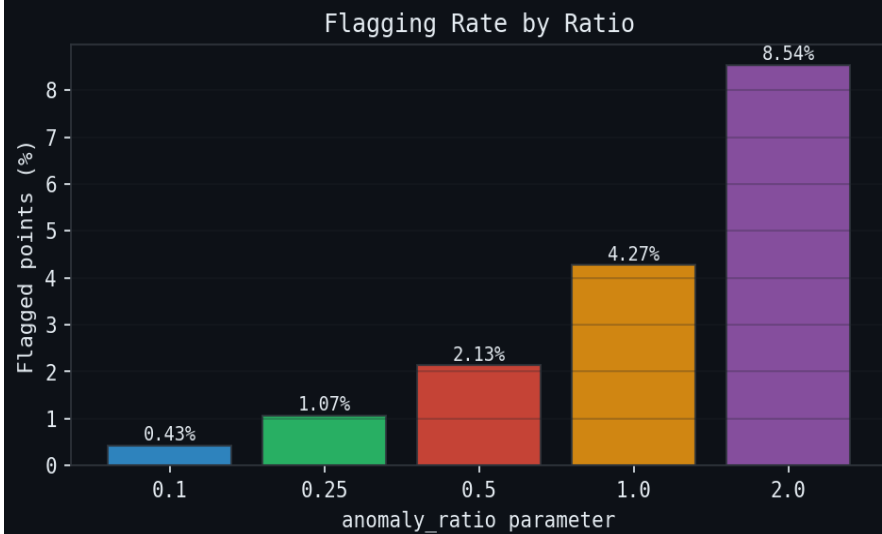
Results by Ratio

anomaly_ratio	Threshold	Flagged Points	Flag Rate	Windows
0.1	0.043692	186	0.426%	151
0.25	0.035213	466	1.067%	330
0.5	0.029050	932	2.134%	573
1.0	0.023646	1,865	4.270%	975
2.0	0.018432	3,730	8.539%	1,586

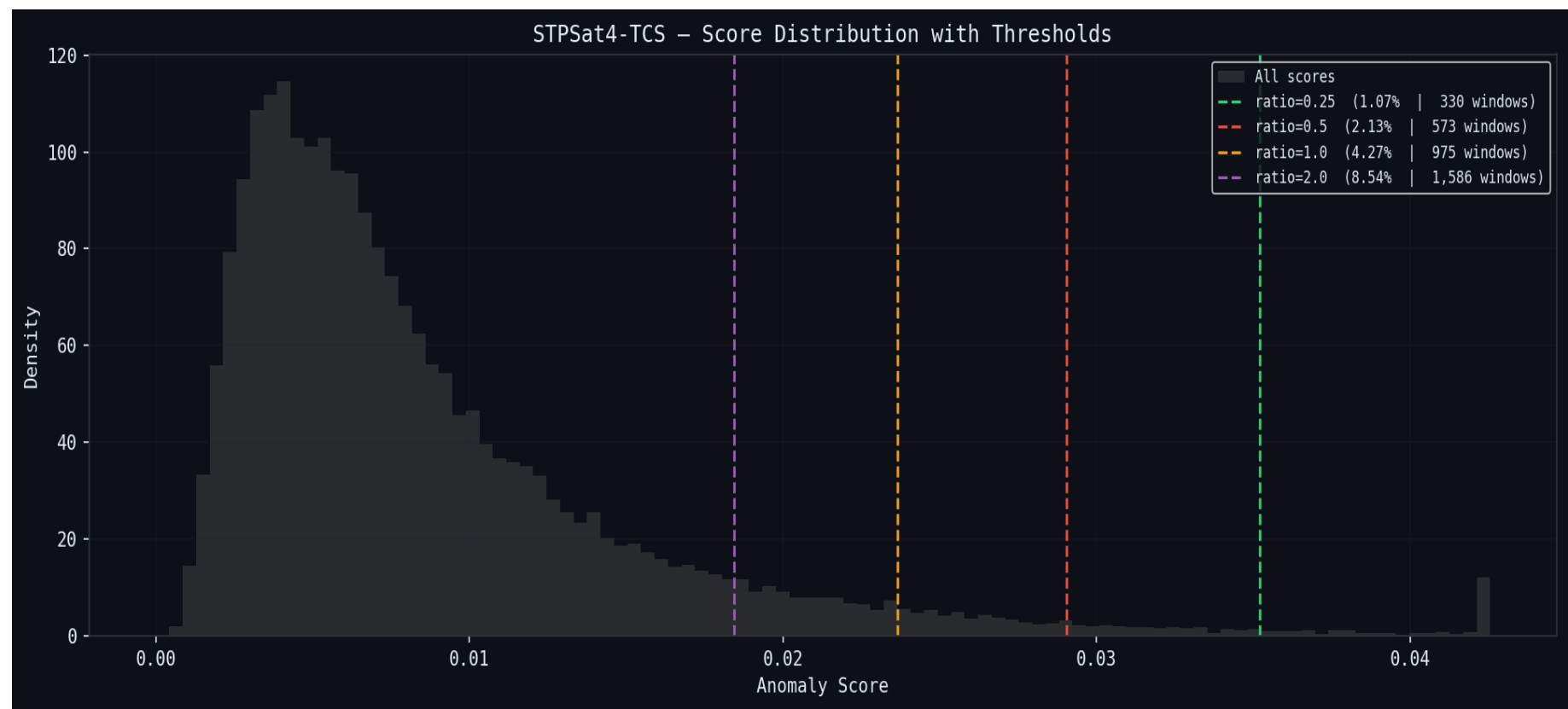
anomaly_ratio controls the detection threshold: UniTS flags the top ratio% of reconstruction error scores as anomalies. Lower values = stricter threshold = fewer flags. The threshold column shows the exact score cutoff for each ratio.

Flagging Rate & Window Count

STPSat4-TCS – Ratio Sweep Summary



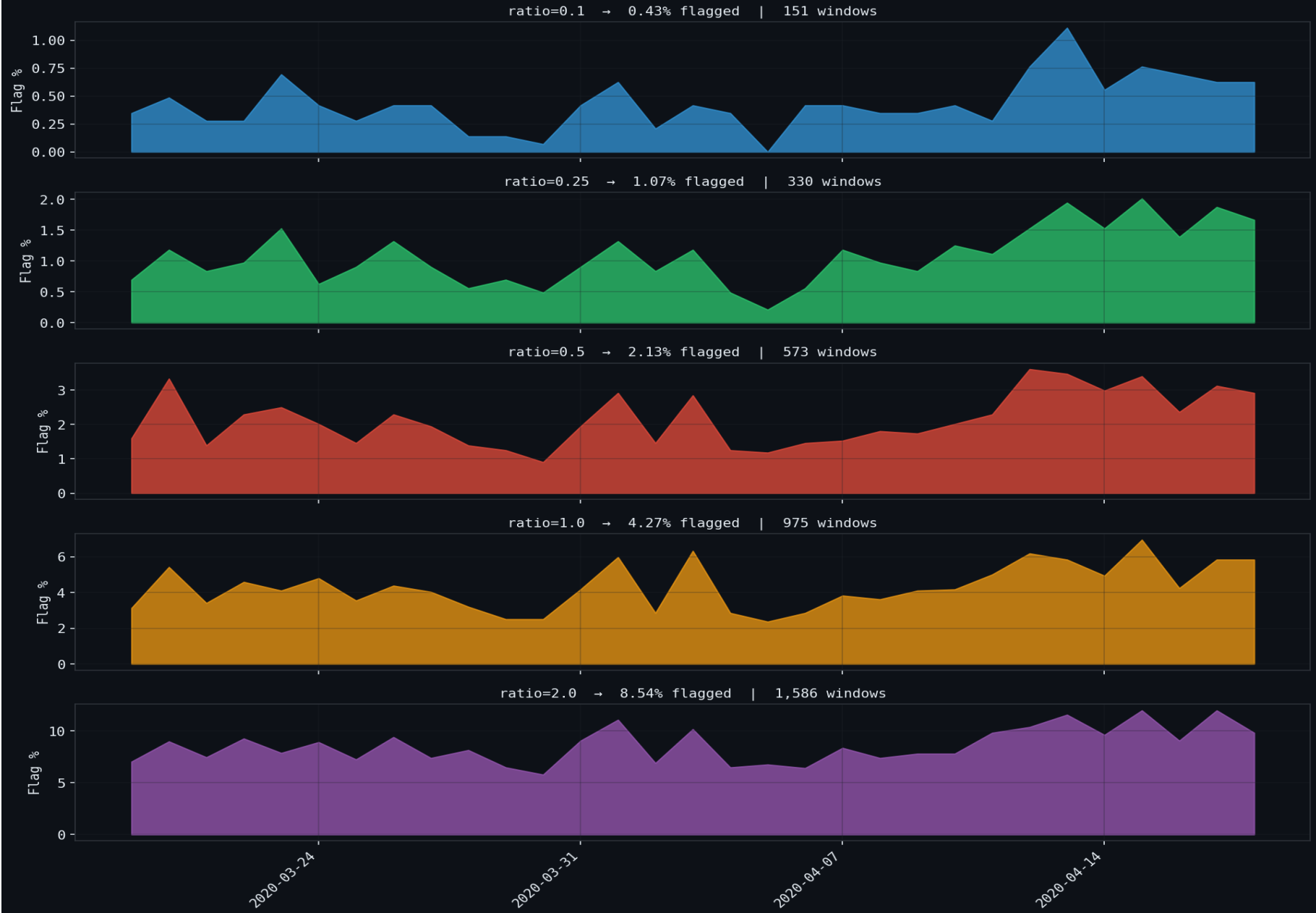
Score Distribution with Thresholds



Dashed vertical lines show where each ratio sets its threshold. Choose a ratio whose threshold sits clearly in the tail of the distribution rather than in the main body of normal scores.

Daily Flagging Rate by Ratio

STPSat4-TCS — Daily Flagging Rate by Ratio



A realistic anomaly rate for a nominal satellite is typically 0.1–1%. Look for a ratio that shows isolated spikes rather than sustained elevated flagging.